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2024 **BORED OF STUDIES TRIAL EXAMINATION**

Mathematics Extension 2

Solutions

Section I

Answers

1	D	**6**	B
2	A	**7**	B
3	C	**8**	A
4	D	**9**	B
5	C	**10**	D

Brief explanations

1 Let θ be the angle between $\underset{\sim}{\hat{a}}$ and $\underset{\sim}{\hat{b}} + \underset{\sim}{\hat{c}}$, then the dot product gives

$$\begin{aligned} \underset{\sim}{\hat{a}} \cdot (\underset{\sim}{\hat{b}} + \underset{\sim}{\hat{c}}) &= |\underset{\sim}{\hat{a}}|\, |\underset{\sim}{\hat{b}} + \underset{\sim}{\hat{c}}| \cos\theta \quad \text{but } |\underset{\sim}{\hat{a}}| = 1 \\ &= \left|\underset{\sim}{\hat{b}} + \underset{\sim}{\hat{c}}\right| \cos\theta \end{aligned}$$

From the triangle inequality, this is maximised if the unit vectors $\underset{\sim}{\hat{b}}$ and $\underset{\sim}{\hat{c}}$ are parallel, which means the maximum value is $|\underset{\sim}{\hat{b}}| + |\underset{\sim}{\hat{c}}|$, which is 2 units.

Hence, the answer is (D).

2 When $x = -2$

$$\begin{aligned} F &= ma \\ &= mv\frac{dv}{dx} \\ &= 4x^3 \\ &= 4(-2)^3 \\ &= -32 \end{aligned}$$

Hence, the answer is (A).

3 Let $a = -5 + 4i, b = 1 + i$ and $c = -1 - 3i$. The complex numbers representing $\overrightarrow{AB}, \overrightarrow{BC}$ and $\overrightarrow{CA}$ are $b - a, c - b$ and $a - c$ respectively.

$$\begin{aligned} b - a &= 6 - 3i \\ c - b &= -2 - 4i \\ a - c &= -4 + 7i \end{aligned}$$

No pair of the vectors have equal length, so it remains to check whether there are any right-angles. First note that $b - a = 3(2 - i)$ and $c - b = -2(1 + 2i)$ so

$$\begin{aligned} i(b - a) &= 3(1 + 2i) \\ &= -\frac{3}{2}(c - b) \end{aligned}$$

This means that there is a rotation of $\frac{\pi}{2}$ between the vectors $\overrightarrow{BA}$ and $\overrightarrow{BC}$.

Hence, the answer is (C).

4 (A) is always true as there will exist any natural number y to ensure that $x + y$ is a natural number.

(B) is always true as there will exist $y = 1$ to ensure that $\frac{x}{y}$ is a natural number.

(C) is always true as all complex numbers have a square root.

(D) is not always true, because for the case when $x = 0$ it is not possible to find any complex number y to ensure $xy = 1$.

Hence, the answer is (D).

5 First note that $z_1 + z_2 = -\frac{b}{a}$ and $z_1 z_2 = \frac{c}{a}$, so

$$\begin{aligned} \arg(abc) &= \arg a + \arg b + \arg c \quad \text{but since } a \text{ is positive real number then } \arg a = 0 \\ &= \arg b + \arg c \\ &= \arg(-a(z_1 + z_2)) + \arg(a z_1 z_2) \\ &= \arg(-a) + \arg(z_1 + z_2) + \arg a + \arg z_1 + \arg z_2 \\ &= \alpha + \beta + \gamma + \pi \end{aligned}$$

Hence, the answer is (C).

6 Since the motion is between $x = -2$ and $x = 2$, then these are the displacement values where $v = 0$. From the acceleration equation

$$\begin{aligned} \ddot{x} &= -x \\ \frac{d}{dx}\left(\frac{v^2}{2}\right) &= -x \\ \frac{v^2}{2} &= -\frac{x^2}{2} + c \qquad \text{when } x = 2, v = 0 \Rightarrow c = 2 \\ \frac{v^2}{2} &= -\frac{x^2}{2} + 2 \\ x^2 + v^2 &= 4 \end{aligned}$$

On the v-x plane this is a circle with the origin as the centre and radius of 2 units. Note that v can be both positive and negative.

Hence, the answer is (B).

7 If the angle between $\underset{\sim}{a}$ and $\underset{\sim}{b} - \underset{\sim}{c}$ is obtuse then its dot product is negative. Also, since $\underset{\sim}{a}$ and $\underset{\sim}{b}$ are perpendicular then $\underset{\sim}{a} \cdot \underset{\sim}{b} = 0$. A necessary condition is then

$$\begin{aligned} \underset{\sim}{a} \cdot (\underset{\sim}{b} - \underset{\sim}{c}) &< 0 \\ \underset{\sim}{a} \cdot \underset{\sim}{b} - \underset{\sim}{a} \cdot \underset{\sim}{c} &< 0 \\ \underset{\sim}{a} \cdot \underset{\sim}{c} &> 0 \end{aligned}$$

For (A), if $\underset{\sim}{c} = -\underset{\sim}{i} - \underset{\sim}{j} - \underset{\sim}{k}$ this requires $-p - q - r > 0$. However, this cannot occur given $0 < p < q < r$.

For (C), if $\underset{\sim}{c} = \underset{\sim}{i} - \underset{\sim}{j} - \underset{\sim}{k}$ this requires $p - q - r > 0$. However, this cannot occur given $p < q$ and $-r < 0$.

For (D), if $\underset{\sim}{c} = -\underset{\sim}{i} + \underset{\sim}{j} - \underset{\sim}{k}$ this requires $-p + q - r > 0$. However, this cannot occur given $q < r$ and $-p < 0$.

For (B), if $\underset{\sim}{c} = \underset{\sim}{i} - \underset{\sim}{j} + \underset{\sim}{k}$ this requires $p - q + r > 0$. This is possible if $r > q - p$ and still follows the condition $0 < p < q < r$.

Hence, the answer is (B).

8 The possible non-real roots of $z^6 = 1$ are

$$\begin{aligned} z &= e^{-i\frac{2\pi}{3}}, e^{-i\frac{\pi}{3}}, e^{i\frac{\pi}{3}}, e^{i\frac{2\pi}{3}} \\ z^2 &= e^{-i\frac{4\pi}{3}}, e^{-i\frac{2\pi}{3}}, e^{i\frac{2\pi}{3}}, e^{i\frac{4\pi}{3}} \\ &= e^{-i\frac{2\pi}{3}}, e^{i\frac{2\pi}{3}} \end{aligned}$$

This means that there are two distinct values for α^2 and β^2. Evaluating their sum and differences gives

$$\begin{aligned} \mathrm{Re}(\alpha^2+\beta^2) &= \mathrm{Re}\left(e^{-i\frac{2\pi}{3}} + e^{i\frac{2\pi}{3}}\right) \\ &= 2\cos\frac{2\pi}{3} < 0 \end{aligned}$$

$$\mathrm{Re}(\alpha^2-\beta^2) = \begin{cases} \mathrm{Re}\left(e^{-i\frac{2\pi}{3}} - e^{i\frac{2\pi}{3}}\right) = 0 \\ \mathrm{Re}\left(e^{i\frac{2\pi}{3}} - e^{-i\frac{2\pi}{3}}\right) = 0 \end{cases}$$

Since $\mathrm{Re}(\alpha^2-\beta^2)=0$, but $\mathrm{Re}(\alpha^2+\beta^2)<0$ always, the answer is (A).

9 By completing the square

$$\begin{aligned} 4x^4-4x^3+x^2+1 &= x^2(4x^2-4x+1)+1 \\ &= x^2(2x-1)^2+1 \\ &= (2x^2-x)^2+1 \\ \int_0^1 \frac{4x-1}{4x^4-4x^3+x^2+1}\,dx &= \int_0^1 \frac{4x-1}{(2x^2-x)^2+1}\,dx \\ &= \left[\tan^{-1}(2x^2-x)\right]_0^1 \\ &= \frac{\pi}{4} \end{aligned}$$

Hence, the answer is (B).

10 Let ω be a root of $P(z)$ and let

$$Q(z) = z^n \cos\theta_n + z^{n-1}\cos\theta_{n-1} + z^{n-2}\cos\theta_{n-2} + \cdots + z\cos\theta_1 + \cos\theta_0.$$

From the triangle inequality and using the fact that $|\cos\theta_k| \leq 1$ for $k = 0, 1, 2, \ldots, n$

$$\begin{aligned}
|Q(\omega)| &= |\omega^n \cos\theta_n + \omega^{n-1}\cos\theta_{n-1} + \omega^{n-2}\cos\theta_{n-2} + \cdots + \omega\cos\theta_1 + \cos\theta_0| \\
&\leq |\omega^n \cos\theta_n| + |\omega^{n-1}\cos\theta_{n-1}| + |\omega^{n-2}\cos\theta_{n-2}| + \cdots + |\omega\cos\theta_1| + |\cos\theta_0| \\
&= |\omega|^n|\cos\theta_n| + |\omega|^{n-1}|\cos\theta_{n-1}| + |\omega|^{n-2}|\cos\theta_{n-2}| + \cdots + |\omega||\cos\theta_1| + |\cos\theta_0| \\
&\leq |\omega|^n + |\omega|^{n-1} + |\omega|^{n-2} + \cdots + |\omega| + 1 \\
&= \frac{1-|\omega|^{n+1}}{1-|\omega|} \quad \text{but } |\omega| > 0 \\
&< \frac{1}{1-|\omega|}
\end{aligned}$$

Since $P(\omega) = 0$ then $Q(\omega) = 2$ so a necessary condition is $\dfrac{1}{1-|\omega|} > 2$.

From (A), if $|\omega| < \dfrac{1}{3}$, then $\dfrac{1}{1-|\omega|} < \dfrac{3}{2}$ which contradicts the necessary condition.

From (B), if $|\omega| < \dfrac{1}{2}$, then $\dfrac{1}{1-|\omega|} < 2$ which contradicts the necessary condition.

Also, note that the contradictions also occur at $|\omega| = \dfrac{1}{3}$ and $|\omega| = \dfrac{1}{2}$.

This implies that the roots must sit in a region that satisfies both $|\omega| > \dfrac{1}{3}$ and $|\omega| > \dfrac{1}{2}$.

Therefore, the smallest region from the choices that contains the roots is $|z| > \dfrac{1}{2}$.

Hence, the answer is (D).

Question 11

(a) Using integration by parts

$u = e^x \tan x \Rightarrow du = e^x(\tan x + \sec^2 x)\, dx$

$dv = \sec x \tan x\, dx \Rightarrow v = \sec x$

$$\begin{aligned}\int e^x \sec x \tan^2 x\, dx &= e^x \sec x \tan x - \int e^x \sec x(\tan x + \sec^2 x)\, dx \\ &= e^x \sec x \tan x - \int e^x \sec x \tan x\, dx - \int e^x \sec^3 x\, dx\end{aligned}$$

Using integration by parts on $\displaystyle\int e^x \sec x \tan x\, dx$

$u = e^x \Rightarrow du = e^x\, dx$

$dv = \sec x \tan x\, dx \Rightarrow v = \sec x$

$$\begin{aligned}\int e^x \sec x \tan^2 x\, dx &= e^x \sec x \tan x - e^x \sec x + \int e^x \sec x\, dx - \int e^x \sec^3 x\, dx \\ &= e^x \sec x \tan x - e^x \sec x - \int e^x \sec x(\sec^2 x - 1)\, dx \\ &= e^x \sec x \tan x - e^x \sec x - \int e^x \sec x \tan^2 x\, dx \\ \int e^x \sec x \tan^2 x\, dx &= \frac{e^x \sec x(\tan x - 1)}{2} + C\end{aligned}$$

(b) In modulus argument form

$$\begin{aligned}1 + i &= \sqrt{2}e^{i\frac{\pi}{4}} \\ \sqrt{3} - i &= 2e^{-i\frac{\pi}{6}}\end{aligned}$$

Since $\dfrac{\pi}{4} - \dfrac{\pi}{6} = \dfrac{\pi}{12}$ then

$$\begin{aligned}(1+i)(\sqrt{3}-i) &= 2\sqrt{2}e^{i\frac{\pi}{12}} \\ \cos\frac{\pi}{12} + i\sin\frac{\pi}{12} &= \frac{(1+i)(\sqrt{3}-i)}{2\sqrt{2}} \\ &= \frac{\sqrt{3} - i + i\sqrt{3} - i^2}{2\sqrt{2}} \\ &= \frac{\sqrt{3} + 1 + i(\sqrt{3} - 1)}{2\sqrt{2}}\end{aligned}$$

Equating real and imaginary parts gives

$$\cos\frac{\pi}{12} = \frac{\sqrt{3}+1}{2\sqrt{2}} \qquad \text{and} \qquad \sin\frac{\pi}{12} = \frac{\sqrt{3}-1}{2\sqrt{2}}$$

(c) (i) Noting that $\overrightarrow{AB}$ is parallel to ℓ_1 and $\overrightarrow{CD}$ is parallel to ℓ_2

$$\begin{aligned}
\overrightarrow{AB} &= \overrightarrow{OB} - \overrightarrow{OA} \\
&= 12\underset{\sim}{i} + 10\underset{\sim}{j} + 5\underset{\sim}{k} - (7\underset{\sim}{i} + 6\underset{\sim}{j} + 2\underset{\sim}{k}) \\
&= 5\underset{\sim}{i} + 4\underset{\sim}{j} + 3\underset{\sim}{k} \\
\overrightarrow{CD} &= \overrightarrow{OD} - \overrightarrow{OC} \\
&= 11\underset{\sim}{i} + 4\underset{\sim}{j} - 2\underset{\sim}{k} - (\underset{\sim}{i} - 4\underset{\sim}{j} - 8\underset{\sim}{k}) \\
&= 10\underset{\sim}{i} + 8\underset{\sim}{j} + 6\underset{\sim}{k} \\
&= 2\overrightarrow{AB}
\end{aligned}$$

Hence, ℓ_1 and ℓ_2 are parallel.

(ii) The vector equation of the line ℓ_1 for some parameter λ is

$$\begin{aligned}
\underset{\sim}{r} &= \overrightarrow{OA} + \lambda\overrightarrow{AB} \\
&= 7\underset{\sim}{i} + 6\underset{\sim}{j} + 2\underset{\sim}{k} + \lambda(5\underset{\sim}{i} + 4\underset{\sim}{j} + 3\underset{\sim}{k})
\end{aligned}$$

The vector equation of the line ℓ_2 for some parameter μ is

$$\begin{aligned}
\underset{\sim}{r} &= \overrightarrow{OC} + \mu\overrightarrow{CD} \\
&= \underset{\sim}{i} - 4\underset{\sim}{j} - 8\underset{\sim}{k} + \mu(10\underset{\sim}{i} + 8\underset{\sim}{j} + 6\underset{\sim}{k})
\end{aligned}$$

(iii) Using the vector equation of ℓ_2 from part (ii)

$$\begin{aligned}
\overrightarrow{AP} &= \overrightarrow{OP} - \overrightarrow{OA} \\
&= (1 + 10\mu)\underset{\sim}{i} + (-4 + 8\mu)\underset{\sim}{j} + (-8 + 6\mu)\underset{\sim}{k} - (7\underset{\sim}{i} + 6\underset{\sim}{j} + 2\underset{\sim}{k}) \\
&= (-6 + 10\mu)\underset{\sim}{i} + (-10 + 8\mu)\underset{\sim}{j} + (-10 + 6\mu)\underset{\sim}{k} \\
\left|\overrightarrow{AP}\right|^2 &= (-6 + 10\mu)^2 + (-10 + 8\mu)^2 + (-10 + 6\mu)^2 \\
&= 36 - 120\mu + 100\mu^2 + 100 - 160\mu + 64\mu^2 + 100 - 120\mu + 36\mu^2 \\
\left|\overrightarrow{AP}\right| &= \sqrt{200\mu^2 - 400\mu + 236}
\end{aligned}$$

(iv) Minimising distance between ℓ_1 and ℓ_2 is equivalent to minimising $|\overrightarrow{AP}|$ or $|\overrightarrow{AP}|^2$.

$$\begin{aligned}
\frac{d}{d\mu}\left|\overrightarrow{AP}\right|^2 &= 400\mu - 400 \\
\frac{d^2}{d\mu^2}\left|\overrightarrow{AP}\right|^2 &= 400 > 0
\end{aligned}$$

Since $\left|\overrightarrow{AP}\right|^2$ is concave up, then it has a minimum value at $\mu = 1$. Since there are no other turning points this is also the global minimum.

Hence, the minimum value of $\left|\overrightarrow{AP}\right|$ is 6 units.

Remark: The vector equation could be equivalently written as

$$\underset{\sim}{r} = \underset{\sim}{i} - 4\underset{\sim}{j} - 8\underset{\sim}{k} + \mu(5\underset{\sim}{i} + 4\underset{\sim}{j} + 3\underset{\sim}{k}).$$

This gives $\left|\overrightarrow{AP}\right| = \sqrt{50\mu^2 - 200\mu + 236}$, which is minimised when $\mu = 2$. This gives the same minimum value of $\left|\overrightarrow{AP}\right|$ of 6 units.

(d) Since $P(x)$ has real coefficients and the roots are all non-real then they must occur in conjugate pairs. This means that the square formed by plotting the roots on the complex plane is bisected by the real axis.

Let the roots be $\alpha, \overline{\alpha}, \beta$, and $\overline{\beta}$. Denote $\alpha = p + iq$ where p and q are real numbers, then $\overline{\alpha} = p - iq$. The distance between the roots α and $\overline{\alpha}$ on the complex plane is $2q$. However, since the roots lie on a square, then the side length of the square must also be $2q$.

The other roots can be derived in terms of p and q. In this case denote $\beta = (p + 2q) + iq$ and $\overline{\beta} = (p + 2q) - iq$.

Taking the sum of the roots.

$$\begin{aligned} p + iq + p - iq + (p + 2q) + iq + (p + 2q) - iq &= 4 \\ 4p + 4q &= 4 \\ p + q &= 1 \end{aligned}$$

Taking the product of the roots and substitute $q = 1 - p$.

$$\begin{aligned} (p + iq)(p - iq)[(p + 2q) + iq][(p + 2q) - iq] &= 5 \\ (p^2 + q^2)[(p + 2q)^2 + q^2] &= 5 \\ [(1 - q)^2 + q^2][(1 + q)^2 + q^2] &= 5 \\ (1 - q)^2(1 + q)^2 + q^2(1 - q)^2 + q^2(1 + q^2) + q^4 &= 5 \\ (1 - q^2)^2 + q^2[(1 - q)^2 + (1 + q)^2] + q^4 &= 5 \\ 1 - 2q^2 + q^4 + 2q^2(1 + q^2) + q^4 &= 5 \\ 4q^4 &= 4 \\ q &= \pm 1 \end{aligned}$$

When $q = -1 \Rightarrow p = 2$, so $\alpha, \overline{\alpha}, \beta$, and $\overline{\beta}$ are $2 - i, 2 + i, -i$ and i respectively.

When $q = 1 \Rightarrow p = 0$, so $\alpha, \overline{\alpha}, \beta$, and $\overline{\beta}$ are $i, -i, 2 + i$ and $2 - i$ respectively.

Note that both cases give the same set of roots.

Hence, the roots of $P(x)$ are $i, -i, 2 + i$ and $2 - i$.

Question 12

(a) Let $u = \dfrac{\pi}{2} - x \Rightarrow du = -dx$. When $x = 0$ then $u = \dfrac{\pi}{2}$ and when $x = \pi$ then $u = -\dfrac{\pi}{2}$.

$$\begin{aligned}\int_0^{\pi} \frac{\sin^3 x}{\sec x + \tan x + \cos x}\,dx &= -\int_{\frac{\pi}{2}}^{-\frac{\pi}{2}} \frac{\sin^3\left(\frac{\pi}{2} - u\right)}{\sec\left(\frac{\pi}{2} - u\right) + \tan\left(\frac{\pi}{2} - u\right) + \cos\left(\frac{\pi}{2} - u\right)}\,du \\ &= \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{\cos^3 u}{\mathrm{cosec} u + \cot u + \sin u}\,du\end{aligned}$$

Let $f(u) = \dfrac{\cos^3 u}{\mathrm{cosec} u + \cot u + \sin u}$, then

$$\begin{aligned}f(-u) &= \frac{\cos^3(-u)}{\mathrm{cosec}(-u) + \cot(-u) + \sin(-u)} \\ &= \frac{\cos^3 u}{-\mathrm{cosec} u - \cot u - \sin u} \\ &= -f(u)\end{aligned}$$

Since $f(u)$ is an odd function, then $\displaystyle\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{\cos^3 u}{\mathrm{cosec} u + \cot u + \sin u}\,du = 0$, hence

$$\int_0^{\pi} \frac{\sin^3 x}{\sec x + \tan x + \cos x}\,dx = 0.$$

(b) Let $z = x + iy$ where x and y are real.

$$\begin{aligned}|e^z - 1| &= 1 \\ \left|e^{x+iy} - 1\right| &= 1 \\ |e^x \cos y - 1 + ie^x \sin y| &= 1 \\ (e^x \cos y - 1)^2 + (e^x \sin y)^2 &= 1 \\ e^{2x}\cos^2 y - 2e^x \cos y + 1 + e^{2x}\sin^2 y &= 1 \\ e^{2x} - 2e^x \cos y &= 0 \\ e^x(e^x - 2\cos y) &= 0 \qquad \text{but } e^x > 0 \\ 2\cos y &= e^x\end{aligned}$$

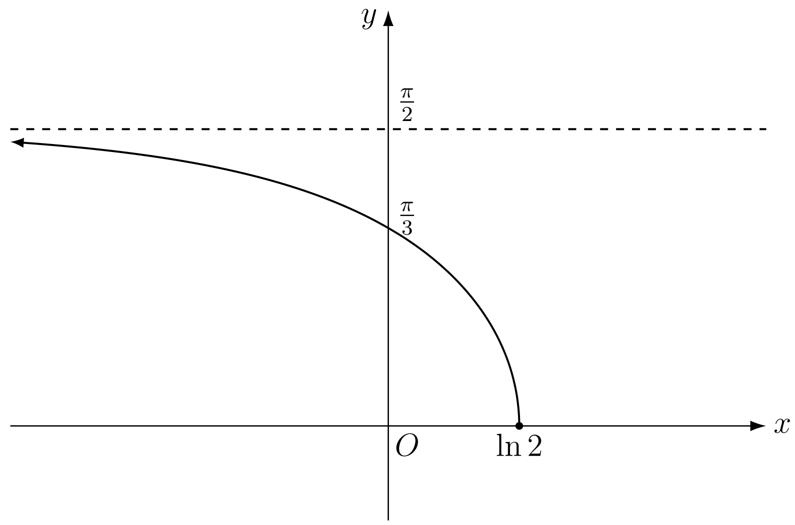
Note that the domain is the solution to $-2 \leq e^x \leq 2$ which gives $x \leq \ln 2$.

Also, the condition $0 \leq \mathrm{Im}(z) < \dfrac{\pi}{2}$ defines the range and ensures there is a unique inverse trigonometric function so that the Cartesian equation is

$$y = \cos^{-1}\left(\frac{e^x}{2}\right)$$

When $x = \ln 2, y = 0$ and as $x \to -\infty, y \to \frac{\pi}{2}$.

Putting this information together gives the following sketch.



(c) (i) The contrapositive statement is

If there does not exist an $x_0 \in [a, b]$ where $f'(x_0) > 0$ then $f(a) \geq f(b)$.

Another way to phrase this is

If $f'(x_0) \leq 0$ for all $x_0 \in [a, b]$ then $f(a) \geq f(b)$.

If $f'(x_0) \leq 0$ for all $x_0 \in [a, b]$ then $f(x)$ is non-increasing in that domain. For continuous and differentiable $f(x)$ this means that $f(a)$ cannot be less than $f(b)$ for $a < b$. Since the contrapositive must be true, then so is the original statement.

(ii) Consider $f(x) = x^2$ in the domain $[-2, 3]$. Since $f(-2) = 4$ and $f(3) = 9$, then $f(-2) < f(3)$. However, $f'(x) = 2x$ which is negative in the interval $[-2, 0)$. This means that the statement is not always true.

(d) (i) From particle P_1's acceleration equation, $n = \dfrac{\pi}{2}$.

Since the centre is the origin, the displacement equation for some constants a (defined as maximum displacement) and α_1 is

$$x_1 = a\cos\left(\frac{\pi t}{2} + \alpha_1\right)$$
$$\dot{x}_1 = -\frac{a\pi}{2}\sin\left(\frac{\pi t}{2} + \alpha_1\right)$$

When $t = 0, x_1 = \dfrac{a}{2}$ and $\dot{x}_1 > 0$ which means

$$\cos\alpha_1 = \frac{1}{2} \quad \text{and} \quad -\frac{a\pi}{2}\sin\alpha_1 > 0$$

This gives $\alpha_1 = -\dfrac{\pi}{3}$. Hence, the displacement equation of P_1 is

$$x_1 = a\cos\left(\frac{\pi t}{2} - \frac{\pi}{3}\right)$$

When $x_1 = a$ then

$$\begin{aligned}\cos\left(\frac{\pi t}{2} - \frac{\pi}{3}\right) &= 1 \\ \frac{\pi t}{2} - \frac{\pi}{3} &= 0, 2\pi, 4\pi, 6\pi, 8\pi, 10\pi, \ldots\end{aligned}$$

The particle is at maximum displacement for the fifth time when

$$\begin{aligned}\frac{\pi t}{2} - \frac{\pi}{3} &= 8\pi \\ t &= \frac{50}{3}\end{aligned}$$

(ii) From particle P_2's acceleration equation, $n = \dfrac{3\pi}{20}$.

Since the centre is the origin, the displacement equation for some constants a (defined as maximum displacement) and α_2 is

$$\begin{aligned}x_2 &= a\cos\left(\frac{3\pi t}{20} + \alpha_2\right) \\ \dot{x}_2 &= -\frac{3a\pi}{20}\sin\left(\frac{3\pi t}{20} + \alpha_2\right)\end{aligned}$$

When $t = 0, x_2 = 0$ and $\dot{x}_2 > 0$ which means

$$\cos\alpha_2 = 0 \quad \text{and} \quad -\frac{3a\pi}{20}\sin\alpha_2 > 0$$

This gives $\alpha_2 = -\dfrac{\pi}{2}$. Hence, the displacement equation of P_2 is

$$x_2 = a\cos\left(\frac{3\pi t}{20} - \frac{\pi}{2}\right)$$

When $x_2 = a$ then

$$\begin{aligned}\cos\left(\frac{3\pi t}{20} - \frac{\pi}{2}\right) &= 1 \\ \frac{3\pi t}{20} - \frac{\pi}{2} &= 0, 2\pi, 4\pi, 6\pi, 8\pi, 10\pi, \ldots \\ \frac{3\pi t}{20} &= \frac{\pi}{2}, \frac{5\pi}{2}, \frac{9\pi}{2}, \frac{13\pi}{2}, \frac{17\pi}{2}, \frac{21\pi}{2}, \ldots \\ t &= \frac{10}{3}, \frac{50}{3}, \frac{90}{3}, \frac{130}{3}, \frac{170}{3}, \frac{210}{3}, \ldots\end{aligned}$$

Since P_1 reaches its maximum displacement for the fifth time when $t = \frac{50}{3}$, this coincides with the second occasion that P_2 reaches its maximum displacement. Hence, $n = 2$.

(e) (i) Let ω be a non-real root of $z^7 = 1$. For integer values of $k = \pm1, \pm2, \pm3$, the root ω has the general form $\omega = \cos\dfrac{2k\pi}{7} + i\sin\dfrac{2k\pi}{7}$.

Let $P(x) = x^3 + x^2 - 2x - 1$. Using the fact that $\omega + \dfrac{1}{\omega} = 2\cos\dfrac{2k\pi}{7}$

$$\begin{aligned}
P\left(2\cos\frac{2k\pi}{7}\right) &= P\left(\omega + \frac{1}{\omega}\right)\\
&= \left(\omega + \frac{1}{\omega}\right)^3 + \left(\omega + \frac{1}{\omega}\right)^2 - 2\left(\omega + \frac{1}{\omega}\right) - 1\\
&= \omega^3 + 3\omega^2\left(\frac{1}{\omega}\right) + 3\omega\left(\frac{1}{\omega^2}\right) + \frac{1}{\omega^3} + \omega^2 + 2 + \frac{1}{\omega^2} - 2\omega - \frac{2}{\omega} - 1\\
&= \omega^3 + \omega + \frac{1}{\omega} + \frac{1}{\omega^3} + \omega^2 + \frac{1}{\omega^2} + 1\\
&= \frac{\omega^6 + \omega^5 + \omega^4 + \omega^3 + \omega^2 + \omega + 1}{\omega^3}\\
&= \frac{\omega^7 - 1}{\omega^3(\omega - 1)} \quad \text{but } \omega^7 = 1 \text{ and } \omega \neq 1 \quad \text{for non-real } \omega\\
&= 0
\end{aligned}$$

However, note that since $\cos(-x) = \cos x$ then there are only just three distinct values of $2\cos\dfrac{2k\pi}{7}$.

Hence, the solutions of $x^3 + x^2 - 2x - 1 = 0$ are $2\cos\dfrac{2\pi}{7}, 2\cos\dfrac{4\pi}{7}$ and $2\cos\dfrac{6\pi}{7}$.

(ii) Taking the product of the roots

$$\begin{aligned}
2\cos\frac{2\pi}{7} \times 2\cos\frac{4\pi}{7} \times 2\cos\frac{6\pi}{7} &= 1\\
\cos\frac{2\pi}{7}\cos\frac{4\pi}{7}\cos\frac{6\pi}{7} &= \frac{1}{8}\\
\sin\left(\frac{\pi}{2} - \frac{2\pi}{7}\right)\sin\left(\frac{\pi}{2} - \frac{4\pi}{7}\right)\sin\left(\frac{\pi}{2} - \frac{6\pi}{7}\right) &= \frac{1}{8}\\
\sin\left(\frac{3\pi}{14}\right)\sin\left(-\frac{\pi}{14}\right)\sin\left(-\frac{5\pi}{14}\right) &= \frac{1}{8}\\
\sin\left(\frac{\pi}{14}\right)\sin\left(\frac{3\pi}{14}\right)\sin\left(\frac{5\pi}{14}\right) &= \frac{1}{8}
\end{aligned}$$

Question 13

(a) Given that $|z_1| = 1$ and $|z_2| = 1$

$$\begin{aligned}
\text{LHS} &= |1+z_1| + |1+z_2| + |1+z_1z_2| \\
&= |1+z_1| + |1+z_2| + |-(1+z_1z_2)| \\
&\geq |1+z_1| + |1+z_2-(1+z_1z_2)| \quad \text{by triangle inequality} \\
&= |1+z_1| + |z_2||1-z_1| \\
&= |1+z_1| + |1-z_1| \\
&\geq |1+z_1+1-z_1| \quad \text{by triangle inequality} \\
&= 2 \\
&= \text{RHS}
\end{aligned}$$

(b) (i) In the upwards trajectory, the object has resistance in the downwards direction. Given $v \geq 0$, it has the net force equation

$$m\ddot{y} = -mg - mkv^2.$$

In the downwards trajectory, the object has resistance in the upwards direction. Given $v \leq 0$, it has the net force equation

$$m\ddot{y} = -mg + mkv^2.$$

This is equivalent to $\ddot{y} = -g - kv|v|$ which accounts for the different cases where $v \geq 0$ and $v \leq 0$.

(ii) For the upwards trajectory, when $y = 0$, $v = u$ and when $y = H$, $v = 0$. The acceleration equation is

$$\begin{aligned}
\ddot{y} &= -g - kv^2 \\
v\frac{dv}{dy} &= -(g+kv^2) \\
\int_0^H dy &= -\int_u^0 \frac{v}{g+kv^2}\,dv \\
[y]_0^H &= -\frac{1}{2k}\left[\ln(g+kv^2)\right]_u^0 \\
H &= \frac{1}{2k}\ln\left(\frac{g+ku^2}{g}\right)
\end{aligned}$$

(iii) For the upwards trajectory, when $t = 0$, $v = u$ and when $t = T_u$, $v = 0$. The acceleration equation is

$$\begin{aligned}
\ddot{y} &= -g - kv^2 \\
\frac{dv}{dt} &= -(g + kv^2) \\
\int_0^{T_u} dt &= -\int_u^0 \frac{dv}{g + kv^2} \\
[t]_0^{T_u} &= -\frac{1}{k}\int_u^0 \frac{dv}{\frac{g}{k} + v^2} \\
T_u &= -\frac{1}{\sqrt{kg}}\left[\tan^{-1}\left(v\sqrt{\frac{k}{g}}\right)\right]_u^0 \\
&= \frac{1}{\sqrt{kg}}\tan^{-1}\left(u\sqrt{\frac{k}{g}}\right)
\end{aligned}$$

(iv) For the downwards trajectory, when $y = H$, $v = 0$ and when $y = 0$, $v = v_g$. The acceleration equation is

$$\begin{aligned}
\ddot{y} &= -g + kv^2 \\
v\frac{dv}{dy} &= -g + kv^2 \\
\int_H^0 dy &= -\int_0^{v_g} \frac{v}{g - kv^2}\,dv \\
[y]_H^0 &= \frac{1}{2k}\left[\ln(g - kv^2)\right]_0^{v_g} \\
-H &= \frac{1}{2k}\ln\left(\frac{g - k{v_g}^2}{g}\right) \\
H &= \frac{1}{2k}\ln\left(\frac{g}{g - k{v_g}^2}\right)
\end{aligned}$$

Equate this with the result in part (ii)

$$\begin{aligned}
\frac{1}{2k}\ln\left(\frac{g}{g - k{v_g}^2}\right) &= \frac{1}{2k}\ln\left(\frac{g + ku^2}{g}\right) \\
\frac{g}{g - k{v_g}^2} &= \frac{g + ku^2}{g} \\
g^2 &= g^2 + kgu^2 - k{v_g}^2(g + ku^2) \\
v_g^2 &= \frac{gu^2}{g + ku^2}
\end{aligned}$$

Remark: Note that the net acceleration of downward trajectory $\ddot{y} < 0$ which implies $g - kv^2 > 0$.

(v) For the downwards trajectory, when $t = T_u$, $v = 0$ and when $t = T_u + T_d$, $v = v_g$. The acceleration equation is

$$\frac{dv}{dt} = -g + kv^2$$

$$\int_{T_u}^{T_u+T_d} dt = -\int_0^{v_g} \frac{dv}{g - kv^2} \qquad \text{note that } g - kv^2 > 0$$

$$[t]_{T_u}^{T_u+T_d} = -\int_0^{v_g} \frac{dv}{(\sqrt{g} - v\sqrt{k})(\sqrt{g} + v\sqrt{k})}$$

$$T_d = -\frac{1}{2\sqrt{g}} \int_0^{v_g} \frac{\sqrt{g} - v\sqrt{k} + \sqrt{g} + v\sqrt{k}}{(\sqrt{g} - v\sqrt{k})(\sqrt{g} + v\sqrt{k})}\, dv$$

$$= -\frac{1}{2\sqrt{g}} \int_0^{v_g} \left(\frac{1}{\sqrt{g} + v\sqrt{k}} + \frac{1}{\sqrt{g} - v\sqrt{k}} \right) dv$$

$$= -\frac{1}{2\sqrt{kg}} \left[\ln\left(\sqrt{g} + v\sqrt{k}\right) - \ln\left(\sqrt{g} - v\sqrt{k}\right) \right]_0^{v_g}$$

$$= -\frac{1}{2\sqrt{kg}} \ln\left(\frac{\sqrt{g} + v_g\sqrt{k}}{\sqrt{g} - v_g\sqrt{k}} \times \frac{\sqrt{g} - v_g\sqrt{k}}{\sqrt{g} - v_g\sqrt{k}} \right)$$

$$= \frac{1}{2\sqrt{kg}} \ln\left(\frac{\left(\sqrt{g} - v_g\sqrt{k}\right)^2}{g - kv_g^2} \right)$$

$$\text{But} \quad g - kv_g^2 = g - \frac{kgu^2}{g + ku^2} \qquad \text{using the result from part (iv)}$$

$$= \frac{g^2}{g + ku^2}$$

$$\text{Also} \quad \sqrt{g} - v_g\sqrt{k} = \sqrt{g} + \frac{u\sqrt{kg}}{\sqrt{g + ku^2}} \qquad \text{since } v_g \leq 0 \text{ in downward trajectory}$$

$$= \frac{\sqrt{g}\left(\sqrt{g + ku^2} + u\sqrt{k}\right)}{\sqrt{g + ku^2}}$$

$$\left(\sqrt{g} - v_g\sqrt{k}\right)^2 = \frac{g\left(\sqrt{g + ku^2} + u\sqrt{k}\right)^2}{g + ku^2}$$

$$\Rightarrow \frac{\left(\sqrt{g} - v_g\sqrt{k}\right)^2}{g - kv_g^2} = \frac{\left(\sqrt{g + ku^2} + u\sqrt{k}\right)^2}{g}$$

$$\text{Hence} \quad T_d = \frac{1}{2\sqrt{kg}} \ln\left(\frac{\left(\sqrt{g + ku^2} + u\sqrt{k}\right)^2}{g} \right)$$

$$T_d = \frac{1}{2\sqrt{kg}} \ln\left(\frac{\sqrt{g + ku^2} + u\sqrt{k}}{\sqrt{g}} \right)^2$$

$$\therefore T_d = \frac{1}{\sqrt{kg}} \ln\left(\frac{\sqrt{g + ku^2} + u\sqrt{k}}{\sqrt{g}} \right)$$

(vi) Let $\omega = u\sqrt{\frac{k}{g}}$ where k and g are positive constants, and $u > 0$

$$\begin{aligned}
T_d - T_u &= \frac{1}{\sqrt{kg}} \ln\left(\frac{\sqrt{g+ku^2}+u\sqrt{k}}{\sqrt{g}}\right) - \frac{1}{\sqrt{kg}} \tan^{-1}\left(u\sqrt{\frac{k}{g}}\right) \\
&= \frac{1}{\sqrt{kg}} \ln\left(\sqrt{1+\frac{ku^2}{g}}+u\sqrt{\frac{k}{g}}\right) - \frac{1}{\sqrt{kg}} \tan^{-1}\left(u\sqrt{\frac{k}{g}}\right) \\
&= \frac{1}{\sqrt{kg}} \ln\left(\sqrt{1+\omega^2}+\omega\right) - \frac{1}{\sqrt{kg}} \tan^{-1}\omega
\end{aligned}$$

Let $f(\omega) = \frac{1}{\sqrt{kg}} \ln\left(\sqrt{1+\omega^2}+\omega\right) - \frac{1}{\sqrt{kg}} \tan^{-1}\omega$

$$\begin{aligned}
f'(\omega) &= \frac{1}{\sqrt{kg}} \times \frac{\frac{\omega}{\sqrt{1+\omega^2}}+1}{\sqrt{1+\omega^2}+\omega} - \frac{1}{\sqrt{kg}} \times \frac{1}{1+\omega^2} \\
&= \frac{w+\sqrt{1+\omega^2}}{\sqrt{kg(1+\omega^2)}(\sqrt{1+\omega^2}+\omega)} - \frac{1}{\sqrt{kg}(1+\omega^2)} \\
&= \frac{1}{\sqrt{kg}}\left(\frac{1}{\sqrt{1+\omega^2}} - \frac{1}{1+\omega^2}\right) \\
&= \frac{1}{\sqrt{kg}}\left(\frac{\sqrt{1+\omega^2}-1}{1+\omega^2}\right)
\end{aligned}$$

When $\omega > 0$ then $\omega^2 > 0 \Rightarrow \sqrt{1+\omega^2} > 1$, hence $f'(\omega) > 0$ for $\omega > 0$.

Since $f(0) = 0$ and $f'(\omega) > 0$ for $\omega > 0$, then it must be that $f(\omega) > 0$ for $\omega > 0$. Hence, $T_d - T_u > 0$, or equivalently $T_d > T_u$.

Question 14

(a) (i) Note that the defined position vectors are all relative to A. The equation of the line through $\overrightarrow{AM}$ for some parameter λ_1 is

$$\begin{aligned} \underset{\sim}{r} &= \lambda_1 \left(\overrightarrow{AB} + \overrightarrow{BM}\right) \\ &= \lambda_1 \left(\underset{\sim}{u} + \frac{1}{2}\underset{\sim}{v}\right) \end{aligned}$$

The equation of the line through $\overrightarrow{BD}$ for some parameter λ_2 is

$$\begin{aligned} \underset{\sim}{r} &= \overrightarrow{AB} + \lambda_2 \left(\overrightarrow{BA} + \overrightarrow{AD}\right) \\ &= \underset{\sim}{u} + \lambda_2 \left(\underset{\sim}{v} - \underset{\sim}{u}\right) \end{aligned}$$

The position of X is the point of intersection of these two lines, so

$$\begin{aligned} \lambda_1 \left(\underset{\sim}{u} + \frac{1}{2}\underset{\sim}{v}\right) &= \underset{\sim}{u} + \lambda_2 \left(\underset{\sim}{v} - \underset{\sim}{u}\right) \\ \lambda_1 \underset{\sim}{u} + \frac{\lambda_1}{2}\underset{\sim}{v} &= \left(1 - \lambda_2\right) \underset{\sim}{u} + \lambda_2 \underset{\sim}{v} \end{aligned}$$

Since $\underset{\sim}{u}$ and $\underset{\sim}{v}$ are perpendicular, the components can be equated.

$$\begin{aligned} \text{Equating } \underset{\sim}{u} \text{ components} \quad & \lambda_1 = 1 - \lambda_2 \Rightarrow \lambda_2 = 1 - \lambda_1 \\ \text{Equating } \underset{\sim}{v} \text{ components} \quad & \frac{\lambda_1}{2} = \lambda_2 \\ & \phantom{\frac{\lambda_1}{2}} = 1 - \lambda_1 \\ & \lambda_1 = \frac{2}{3} \quad \text{and} \quad \lambda_2 = \frac{1}{3} \end{aligned}$$

Substitute into equation of line passing through $\overrightarrow{AM}$ to get position vector $\overrightarrow{AX}$

$$\begin{aligned} \underset{\sim}{r} &= \frac{2}{3}\left(\underset{\sim}{u} + \frac{1}{2}\underset{\sim}{v}\right) \\ \overrightarrow{AX} &= \frac{2}{3}\underset{\sim}{u} + \frac{1}{3}\underset{\sim}{v} \end{aligned}$$

(ii) Given that $\overrightarrow{BD} = \underset{\sim}{v} - \underset{\sim}{u}$

$$\begin{aligned} \overrightarrow{BX} &= \overrightarrow{AX} - \overrightarrow{AB} \\ &= \frac{2}{3}\underset{\sim}{u} + \frac{1}{3}\underset{\sim}{v} - \underset{\sim}{u} \\ &= \frac{1}{3}(\underset{\sim}{v} - \underset{\sim}{u}) \\ &= \frac{1}{3}\overrightarrow{BD} \\ \left|\overrightarrow{BX}\right| &= \frac{1}{3}\left|\overrightarrow{BD}\right| \\ &= \frac{1}{3}\left(\left|\overrightarrow{BX}\right| + \left|\overrightarrow{XD}\right|\right) \\ \left|\overrightarrow{XD}\right| &= 2\left|\overrightarrow{BX}\right| \end{aligned}$$

(iii) The equation of the line through $\overrightarrow{PC}$ for some parameter λ_3 is

$$\begin{aligned}
\underset{\sim}{r} &= \overrightarrow{AP} + \lambda_3 \left(\overrightarrow{PB} + \overrightarrow{BC}\right) \\
&= \frac{1}{2}\underset{\sim}{u} + \lambda_3 \left(\frac{1}{2}\underset{\sim}{u} + \underset{\sim}{v}\right) \\
&= \frac{\lambda_3 + 1}{2}\,\underset{\sim}{u} + \lambda_3\,\underset{\sim}{v}
\end{aligned}$$

If X lies on this line then there should exist a value of λ_3 where $\underset{\sim}{r} = \overrightarrow{AX}$.

Note that if $\lambda_3 = \dfrac{1}{3}$ then $\dfrac{\lambda_3 + 1}{2} = \dfrac{2}{3} \Rightarrow \underset{\sim}{r} = \dfrac{2}{3}\underset{\sim}{u} + \dfrac{1}{3}\underset{\sim}{v}$,

Hence, X lies on PC.

(b) (i) Find the vectors of each side in terms of $\underset{\sim}{u}, \underset{\sim}{v}$ and $\underset{\sim}{w}$

$$\begin{aligned}
\overrightarrow{AM} &= \overrightarrow{AB} + \overrightarrow{BM} \\
&= \underset{\sim}{u} + \frac{1}{2}\underset{\sim}{v} \\
\overrightarrow{NH} &= \overrightarrow{NE} + \overrightarrow{EH} \\
&= \frac{1}{2}\underset{\sim}{v} + \underset{\sim}{u} \\
\overrightarrow{MH} &= \overrightarrow{MC} + \overrightarrow{CH} \\
&= \frac{1}{2}\underset{\sim}{v} + \underset{\sim}{w} \\
\overrightarrow{AN} &= \overrightarrow{AF} + \overrightarrow{FN} \\
&= \underset{\sim}{w} + \frac{1}{2}\underset{\sim}{v}
\end{aligned}$$

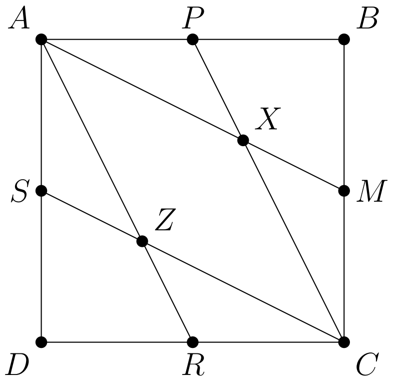
Since, $\overrightarrow{AM} = \overrightarrow{NH}$ and $\overrightarrow{MH} = \overrightarrow{AN}$ then opposite sides are parallel. Also,

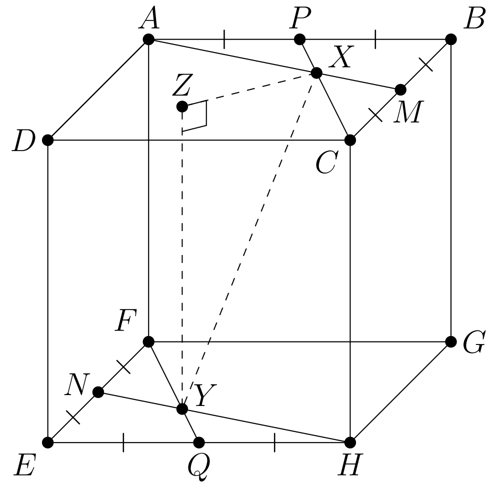
$$\begin{aligned}
\left|\overrightarrow{AM}\right| &= \sqrt{\left|\overrightarrow{AB}\right|^2 + \left|\overrightarrow{BM}\right|^2} \\
&= \sqrt{5} \\
\left|\overrightarrow{MH}\right| &= \sqrt{\left|\overrightarrow{CM}\right|^2 + \left|\overrightarrow{CH}\right|^2} \\
&= \sqrt{5}
\end{aligned}$$

Since opposite sides are parallel and adjacent sides are equal then $AMHN$ is a rhombus.

(ii) Let S and R be the midpoints of AD and DC respectively. Let Z be the intersection of AR and SC.

By symmetry of the cube, Z is directly positioned above Y, so $\left|\overrightarrow{YZ}\right| = 2$.





Using the result from (a)(ii) and symmetry of the square

$$\begin{aligned}\left|\overrightarrow{BX}\right| &= \frac{1}{3}\left|\overrightarrow{BD}\right|\\ \left|\overrightarrow{DZ}\right| &= \frac{1}{3}\left|\overrightarrow{BD}\right|\end{aligned}$$

From (a)(iii), the diagonal $\overrightarrow{BD}$ passes through X and Z so

$$\begin{aligned}\left|\overrightarrow{XZ}\right| &= \left|\overrightarrow{BD}\right| - \left|\overrightarrow{BX}\right| - \left|\overrightarrow{DZ}\right|\\ &= \frac{1}{3}\left|\overrightarrow{BD}\right|\\ &= \frac{1}{3}\sqrt{\left|\overrightarrow{AB}\right|^2 + \left|\overrightarrow{AD}\right|^2}\\ &= \frac{2\sqrt{2}}{3}\\ \left|\overrightarrow{XY}\right|^2 &= \left|\overrightarrow{XZ}\right|^2 + \left|\overrightarrow{YZ}\right|^2\\ &= \frac{8}{9} + 4\\ \left|\overrightarrow{XY}\right| &= \frac{2\sqrt{11}}{3}\end{aligned}$$

(c) (i) Consider

$$\begin{aligned}
I_n + aI_{n-1} &= \int_0^1 \frac{x^n}{x+a}\,dx + \int_0^1 \frac{ax^{n-1}}{x+a}\,dx \\
&= \int_0^1 \frac{x^{n-1}(x+a)}{x+a}\,dx \\
&= \int_0^1 x^{n-1}\,dx \\
&= \left[\frac{x^n}{n}\right]_0^1 \\
&= \frac{1}{n}
\end{aligned}$$

$$\therefore I_n = \frac{1}{n} - aI_{n-1}$$

(ii) Applying the recurrence relations repeatedly

$$\begin{aligned}
I_n &= \frac{1}{n} - aI_{n-1} \\
&= \frac{1}{n} - a\left(\frac{1}{n-1} - aI_{n-2}\right) \\
&= \frac{1}{n} - \frac{a}{n-1} + a^2\left(\frac{1}{n-2} - aI_{n-3}\right) \\
&= \frac{1}{n} - \frac{a}{n-1} + \frac{a^2}{n-2} - a^3\left(\frac{1}{n-3} - aI_{n-4}\right) \\
&= \frac{1}{n} - \frac{a}{n-1} + \frac{a^2}{n-2} - \frac{a^3}{n-3} + a^4\left(\frac{1}{n-4} - aI_{n-5}\right) \\
&\vdots \\
&= \frac{1}{n} - \frac{a}{n-1} + \frac{a^2}{n-2} - \frac{a^3}{n-3} + \cdots + \frac{(-1)^{n-2}a^{n-2}}{2} + (-1)^{n-1}a^{n-1}(1 - aI_0) \\
&= \sum_{k=0}^{n-1} \frac{(-1)^k a^k}{n-k} + (-1)^n a^n \int_0^1 \frac{dx}{x+a} \\
&= \sum_{k=0}^{n-1} \frac{(-1)^k a^k}{n-k} + (-1)^n a^n \left[\ln(x+a)\right]_0^1 \\
&= \sum_{k=0}^{n-1} \frac{(-1)^k a^k}{n-k} + (-1)^n a^n \ln\left(\frac{a+1}{a}\right) \\
&= \sum_{k=0}^{n-1} \frac{(-1)^k a^k}{n-k} + (-1)^n a^n \ln\left(1 + \frac{1}{a}\right)
\end{aligned}$$

(iii) For some constants a and b, let

$$\frac{1}{(x+1)(2x+1)} = \frac{a}{x+1} + \frac{b}{2x+1}$$
$$a(2x+1) + b(x+1) = 1$$

When $x = -1 \Rightarrow a = -1$ and when $x = -\frac{1}{2} \Rightarrow b = 2$, hence

$$\frac{1}{(x+1)(2x+1)} = -\frac{1}{x+1} + \frac{2}{2x+1}.$$

$$\begin{aligned}\int_0^1 \frac{x^4}{(x+1)(2x+1)}\,dx &= -\int_0^1 \frac{x^4}{x+1}\,dx + \int_0^1 \frac{2x^4}{2x+1}\,dx \\ &= \int_0^1 \frac{x^4}{x+\frac{1}{2}}\,dx - \int_0^1 \frac{x^4}{x+1}\,dx\end{aligned}$$

Using the result in part (ii)

$$\begin{aligned}\int_0^1 \frac{x^4}{x+\frac{1}{2}}\,dx &= \sum_{k=0}^{3} \frac{(-1)^k \left(\frac{1}{2}\right)^k}{4-k} + (-1)^4 \left(\frac{1}{2}\right)^4 \ln\left(1 + \frac{1}{\frac{1}{2}}\right) \\ &= \sum_{k=0}^{3} \frac{(-1)^k \left(\frac{1}{2}\right)^k}{4-k} + \frac{1}{16}\ln 3 \\ &= \left(\frac{1}{4} - \frac{\frac{1}{2}}{3} + \frac{\frac{1}{2^2}}{2} - \frac{\frac{1}{2^3}}{1}\right) + \frac{1}{16}\ln 3 \\ &= \frac{1}{12} + \frac{1}{16}\ln 3 \\ \int_0^1 \frac{x^4}{x+1}\,dx &= \sum_{k=0}^{3} \frac{(-1)^k 1^k}{4-k} + (-1)^4 1^4 \ln\left(1 + \frac{1}{1}\right) \\ &= \sum_{k=0}^{3} \frac{(-1)^k}{4-k} + \ln 2 \\ &= \left(\frac{1}{4} - \frac{1}{3} + \frac{1}{2} - \frac{1}{1}\right) + \ln 2 \\ &= -\frac{7}{12} + \ln 2\end{aligned}$$

Hence

$$\begin{aligned}\int_0^1 \frac{x^4}{(x+1)(2x+1)}\,dx &= \frac{1}{12} + \frac{1}{16}\ln 3 + \frac{7}{12} - \ln 2 \\ &= \frac{2}{3} - \ln 2 + \frac{1}{16}\ln 3\end{aligned}$$

Question 15

(a) (i) Let (x, y, z) be a general point on the plane. Since $(0, 0, -1)$ lies on the plane and the position vector $\underset{\sim}{i} - \underset{\sim}{k}$ is perpendicular to the plane then

$$\begin{aligned}(\underset{\sim}{i} - \underset{\sim}{k}) \cdot ((x-0)\underset{\sim}{i} + (y-0)\underset{\sim}{j} + (z+1)\underset{\sim}{k}) &= 0 \\ x - (z+1) &= 0 \\ x - z &= 1\end{aligned}$$

(ii) The parametric coordinates of $x^2 + y^2 = 8$ are

$$x = 2\sqrt{2}\cos\theta \qquad \text{and} \qquad y = 2\sqrt{2}\sin\theta$$

Where the cylinder intersects with $x - z = 1$, gives the parametric coordinates of z for $\mathcal{C}$, so $z = 2\sqrt{2}\sin\theta - 1$. Hence, the parametric coordinates of $\mathcal{C}$ are

$$\left(2\sqrt{2}\cos\theta, 2\sqrt{2}\sin\theta, 2\sqrt{2}\sin\theta - 1\right)$$

(iii) At the point $(2, 2, 1)$ on $\mathcal{C}$, a value of θ can be derived

$$\begin{aligned}2\sqrt{2}\cos\theta &= 2 \\ \cos\theta &= \frac{1}{\sqrt{2}} \\ \theta &= \frac{\pi}{4}\end{aligned}$$

The direction vector at $\theta = \dfrac{\pi}{4}$ is

$$\begin{aligned}\underset{\sim}{r}_d &= -2\sqrt{2}\sin\theta\underset{\sim}{i} + 2\sqrt{2}\cos\theta\underset{\sim}{j} + 2\sqrt{2}\cos\theta\underset{\sim}{k} \\ &= -2\underset{\sim}{i} + 2\underset{\sim}{j} + 2\underset{\sim}{k} \\ &= 2(-\underset{\sim}{i} + \underset{\sim}{j} + \underset{\sim}{k})\end{aligned}$$

Hence, the vector equation of the tangent is

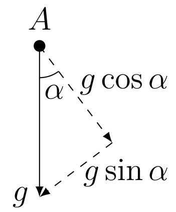
$$\begin{aligned}\underset{\sim}{r} &= 2\underset{\sim}{i} + 2\underset{\sim}{j} + \underset{\sim}{k} + \mu(-2\underset{\sim}{i} + 2\underset{\sim}{j} + 2\underset{\sim}{k}) \\ \text{or equivalently} \quad \underset{\sim}{r} &= 2\underset{\sim}{i} + 2\underset{\sim}{j} + \underset{\sim}{k} + \lambda(-\underset{\sim}{i} + \underset{\sim}{j} + \underset{\sim}{k})\end{aligned}$$

for some parameters μ and λ.

(b) Let (x_c, y_c) be the point that the object lands on the inclined plane, which occurs when $t = T$, where $T \neq 0$. Substitute this into the displacement components.

$$\begin{aligned}
x_c &= v_0 T \cos\theta \\
y_c &= v_0 T \sin\theta - \frac{gT^2}{2} \\
\tan\alpha &= \frac{y_c}{x_c} \\
&= \frac{v_0 T \sin\theta - \frac{gT^2}{2}}{v_0 T \cos\theta} \\
v_0 \cos\theta \tan\alpha &= v_0 \sin\theta - \frac{gT}{2} \\
T &= \frac{2v_0}{g}(\sin\theta - \cos\theta\tan\alpha) \\
&= \frac{2v_0\cos\theta}{g}(\tan\theta - \tan\alpha)
\end{aligned}$$

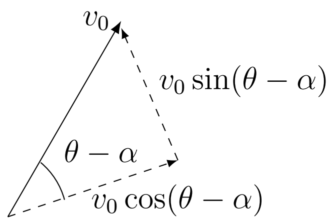
(c) (i) Resolving the gravitational acceleration vector in the $\underset{\sim}{i}$ and $\underset{\sim}{j}$ directions.



The net acceleration on A is

$$\ddot{\underset{\sim}{r}}_A = -g\sin\alpha\,\underset{\sim}{i} - g\cos\alpha\,\underset{\sim}{j}.$$

Resolving the initial velocity vector into the $\underset{\sim}{i}$ and $\underset{\sim}{j}$ components.



So when $t = 0$ the $\underset{\sim}{i}$ and $\underset{\sim}{j}$ components of the velocity vector are $v_0\cos(\theta-\alpha)$ and $v_0\sin(\theta-\alpha)$ respectively. Hence

$$\dot{\underset{\sim}{r}}_A = (-gt\sin\alpha + v_0\cos(\theta-\alpha))\,\underset{\sim}{i} + (-gt\cos\alpha + v_0\sin(\theta-\alpha))\,\underset{\sim}{j}$$

When $t = 0$, the displacement vector is $0\underset{\sim}{i} + 0\underset{\sim}{j}$ so

$$\underset{\sim}{r}_A = \left(v_0 t\cos(\theta-\alpha) - \frac{gt^2\sin\alpha}{2}\right)\underset{\sim}{i} + \left(v_0 t\sin(\theta-\alpha) - \frac{gt^2\cos\alpha}{2}\right)\underset{\sim}{j}.$$

(ii) Since object B moves parallel to the direction of $\underset{\sim}{i}$, then its $\underset{\sim}{j}$ component is always zero. By similarly resolving the gravitational force, its acceleration vector is

$$\ddot{\underset{\sim}{r}}_B = -g \sin\alpha \, \underset{\sim}{i}$$

When $t = 0$, its initial velocity in the $\underset{\sim}{i}$ direction has magnitude u_0, so

$$\dot{\underset{\sim}{r}}_B = (-gt\sin\alpha + u_0)\, \underset{\sim}{i}$$

When $t = 0$, the displacement vector is $0\underset{\sim}{i} + 0\underset{\sim}{j}$ so

$$\underset{\sim}{r}_B = \left(u_0 t - \frac{gt^2 \sin\alpha}{2}\right) \underset{\sim}{i}$$

(iii) The maximum of $\underset{\sim}{r}_B$ occurs when $\dot{\underset{\sim}{r}}_B = 0\underset{\sim}{i}$.

$$\begin{aligned} -gt\sin\alpha + u_0 &= 0 \\ t &= \frac{u_0}{g\sin\alpha} \end{aligned}$$

Equate this with T from part (b) when object A and object B collide.

$$\begin{aligned} \frac{u_0}{g\sin\alpha} &= \frac{2v_0\cos\theta}{g}(\tan\theta - \tan\alpha) \\ u_0 &= 2v_0\cos\theta\sin\alpha(\tan\theta - \tan\alpha) \\ &= 2v_0\cos\theta\sin\alpha\left(\frac{\sin\theta}{\cos\theta} - \frac{\sin\alpha}{\cos\alpha}\right) \\ &= 2v_0\cos\theta\sin\alpha\left(\frac{\sin\theta\cos\alpha - \sin\alpha\cos\theta}{\cos\alpha\cos\theta}\right) \\ &= 2v_0\tan\alpha\sin(\theta - \alpha) \end{aligned}$$

When objects A and B collide, their displacements must be equal. So equating the $\underset{\sim}{i}$ components of $\underset{\sim}{r}_A$ and $\underset{\sim}{r}_B$ gives

$$\begin{aligned} v_0 t\cos(\theta - \alpha) - \frac{gt^2\sin\alpha}{2} &= u_0 t - \frac{gt^2\sin\alpha}{2} \qquad \text{noting } t \neq 0 \\ u_0 &= v_0\cos(\theta - \alpha) \end{aligned}$$

Equating this with the earlier expression for u_0 gives

$$\begin{aligned} v_0\cos(\theta - \alpha) &= 2v_0\tan\alpha\sin(\theta - \alpha) \\ \tan(\theta - \alpha)\tan\alpha &= \frac{1}{2} \end{aligned}$$

Remark: Note that when object A hits the inclined plane, its $\underset{\sim}{j}$ component is zero. When applying this to solve for t, the same time of flight result as part (b) is derived. The equations lead to similar results regardless of whether the reference coordinates are horizontal/vertical or parallel/perpendicular to the inclined plane.

Question 16

(a) (i) When $n = 1$, it is given that w_2 is divisible by 11 so $a_0 - a_1 = 11M$ where $M \in \mathbb{Z}$.

$$\begin{aligned} \text{Consider} \quad u_2 &= a_0 + 10a_1 \\ &= 11M + a_1 + 10a_1 \\ &= 11(M + a_1) \\ &= 11N \quad \text{where } N = M + a_1 \in \mathbb{Z} \end{aligned}$$

The statement is true for $n = 1$.

Assume that the statement is true for $n = k$, that is for integer Q,

$$u_{2k} = 11Q.$$

Required to prove statement is true for $n = k + 1$, that is for integer S,

$$u_{2k+2} = 11S.$$

It is given that $w_{2k} = 11P$ and $w_{2k+2} = 11R$ for integers P and R, so

$$\begin{aligned} w_{2k+2} &= \sum_{k=0}^{2k+1} (-1)^j a_j \\ 11R &= \sum_{k=0}^{2k-1} (-1)^j a_j + a_{2k} - a_{2k+1} \\ &= 11P + a_{2k} - a_{2k+1} \\ a_{2k} - a_{2k+1} &= 11(R - P) \end{aligned}$$

Now consider

$$\begin{aligned} u_{2k+2} &= \sum_{k=0}^{2k+1} 10^j a_j \\ &= \sum_{k=0}^{2k-1} 10^j a_j + 10^{2k} a_{2k} + 10^{2k+1} a_{2k+1} \\ &= 11Q + 10^{2k} a_{2k} + 10^{2k+1} a_{2k+1} \quad \text{by assumption} \\ &= 11Q + 10^{2k}(a_{2k} + 10a_{2k+1}) \\ &= 11Q + 10^{2k}(a_{2k+1} + 11(R - P) + 10a_{2k+1}) \\ &= 11Q + 11.10^{2k}(a_{2k+1} + R - P) \\ &= 11[Q + 10^{2k}(a_{2k+1} + R - P)] \\ &= 11S \quad \text{where } S = Q + 10^{2k}(a_{2k+1} + R - P) \in \mathbb{Z} \end{aligned}$$

Since the statement is true for $n = 1$, then by induction it is true for all positive integers n.

(ii) If u_{2n} is palindromic number then $a_0 = a_{2n-1}, a_1 = a_{2n-2}, \dots, a_{n-1} = a_n$. This implies that

$$\begin{aligned} w_{2n} &= (a_0 - a_{2n-1}) + (-a_1 + a^{2n-2}) + (a_2 - a_{2n-3}) + \cdots + ((-1)^{n-1}a_{n-1} + (-1)^n a_n) \\ &= 0 \end{aligned}$$

Since w_{2n} is divisible by 11 then from part (i), u_{2n} is also divisible by 11. Hence, all palindromic numbers with an even number of digits is divisible by 11.

(b) For $x \in [a, b]$, noting that $f(x) > 0$ and $f'(x) > 0$

$$\begin{aligned} \left(\sqrt{f'(x)} - \frac{1}{\sqrt{f'(x)}}\right)^2 &\geq 0 \\ f'(x) + \frac{1}{f'(x)} &\geq 2 \\ f(x)f'(x) + \frac{f(x)}{f'(x)} &\geq 2f(x) \\ \int_a^b f(x)f'(x)\,dx + \int_a^b \frac{f(x)}{f'(x)}\,dx &\geq 2\int_a^b f(x)\,dx \\ \int_a^b \frac{f(x)}{f'(x)}\,dx &\geq 2[F(x)]_a^b - \frac{1}{2}\left[(f(x))^2\right]_a^b \\ &= 2[F(b) - F(a)] + \frac{1}{2}[(f(a))^2 - (f(b))^2] \\ &= 2[F(b) - F(a)] + \frac{1}{2}[f(a) - f(b)][f(a) + f(b)] \end{aligned}$$

(c) (i) Given $\pi = \dfrac{p}{q}$

$$\begin{aligned} \text{LHS} &= f(\pi - x) \\ &= f\left(\frac{p}{q} - x\right) \\ &= \frac{\left(\frac{p}{q} - x\right)^n \left(p - q\left(\frac{p}{q} - x\right)\right)^n}{n!} \\ &= \frac{1}{n!}\left(\frac{p - qx}{q}\right)^n (p - p + qx)^n \\ &= \frac{x^n(p - qx)^n}{n!} \\ &= f(x) \\ &= \text{RHS} \end{aligned}$$

(ii) Firstly, let $q(x) = x^k$ for sufficiently large positive integer k. By considering the multiple derivatives

$$\begin{aligned}
q^{(1)}(x) &= kx^{k-1} \\
q^{(2)}(x) &= k(k-1)x^{k-2} \\
q^{(3)}(x) &= k(k-1)(k-2)x^{k-3} \\
&\vdots \\
q^{(k-1)}(x) &= k(k-1)(k-2)\ldots 3.2x^1 \\
q^{(k)}(x) &= k(k-1)(k-2)\ldots 3.2.1 \\
q^{(k+1)}(x) &= 0.
\end{aligned}$$

This means that $q^{(1)}(0), q^{(2)}(0), q^{(3)}(0), \ldots, q^{(k-1)}(0), q^{(k+1)}(0)$ all equal zero, which implies that for some positive integer m, $q^{(m)}(0) = 0$ if $m \neq k$. The only non-zero derivative at $x = 0$ is when $m = k$, which gives $q^{(k)}(0) = k!$.

Now the above result will be applied on the derivatives of $f(x)$, which are in $g(x)$. First note that with $f(0) = 0$ then

$$g(0) = \sum_{k=1}^{n} (-1)^k f^{(2k)}(0).$$

Now $x^n(p - qx)^n$ can be written generally as a polynomial of degree $2n$ when the binomial is expanded. Let the coefficient of x^k be a_k, which are integers (since p and q are integers), then

$$\begin{aligned}
f(x) &= \frac{x^n(p-qx)^n}{n!} \\
&= \frac{1}{n!}(a_{2n}x^{2n} + a_{2n-1}x^{2n-1} + a_{2n-2}x^{2n-2} + \cdots + a_n x^n)
\end{aligned}$$

First consider the individual $f^{(2k)}(0)$ terms in $g(x)$ where $2k < n$. Since all terms in $f(x)$ have integer powers of x of at least n, then $f^{(2k)}(0) = 0$.

Secondly, consider the individual $f^{(2k)}(0)$ terms in $g(x)$ where $n \leq 2k \leq 2n$. Each of the individual terms of $f^{(2k)}(0)$ will vanish at $x = 0$, except for one term which will be constant and has a factor of $n!$. For example,

$$\begin{aligned}
f^{(2n-2)}(0) &= \frac{1}{n!}(a_{2n-2}(2n-2)!) \\
&= \frac{a_{2n-2}(2n-2)(2n-3)(2n-4)\ldots(n+2)(n+1)n!}{n!} \\
&= a_{2n-2}(2n-2)(2n-3)\ldots(n+2)(n+1) \quad \text{which is an integer}
\end{aligned}$$

Applying this similarly to each $f^{(2k)}(0)$ term for $n \leq 2k \leq 2n$ will return an integer. Hence, it must be that $g(0)$ is an integer.

Now consider $g(\pi)$. From part (i),

$$\begin{aligned}
f(\pi - x) &= f(x) \\
-f^{(1)}(\pi - x) &= f^{(1)}(x) \\
f^{(2)}(\pi - x) &= f^{(2)}(x) \\
-f^{(3)}(\pi - x) &= f^{(3)}(x) \\
&\vdots \\
(-1)^{2k} f^{(2k)}(\pi - x) &= f^{(2k)}(x)
\end{aligned}$$

Substituting $x = \pi$, gives $f(0) = f(\pi)$ and $f^{(2k)}(0) = f^{(2k)}(\pi)$ so

$$\begin{aligned}
g(\pi) &= f(\pi) + \sum_{k=1}^{n} (-1)^k f^{(2k)}(\pi) \\
&= f(0) + \sum_{k=1}^{n} (-1)^k f^{(2k)}(0) \\
&= g(0) \quad \text{which was shown to be an integer}
\end{aligned}$$

Hence, $g(0) + g(\pi)$ is an integer.

(iii)

$$\begin{aligned}
\frac{d}{dx}\left[g'(x)\sin x - g(x)\cos x\right] &= g''(x)\sin x + g'(x)\cos x - g'(x)\cos x + g(x)\sin x \\
&= (g''(x) + g(x))\sin x
\end{aligned}$$

But

$$\begin{aligned}
g(x) &= f(x) + \sum_{k=1}^{n} (-1)^k f^{(2k)}(x) \\
&= f(x) - f^{(2)}(x) + f^{(4)}(x) - f^{(6)}(x) + \cdots + (-1)^{n-1} f^{(2n-2)}(x) + (-1)^n f^{(2n)}(x) \\
g''(x) &= f^{(2)}(x) - f^{(4)}(x) + f^{(6)}(x) - f^{(8)}(x) + \cdots + (-1)^{n-1} f^{(2n)}(x) + (-1)^n f^{(2n+2)}(x)
\end{aligned}$$

Noting that $f(x)$ has x^{2n} as the term with the largest power

$$\begin{aligned}
g''(x) + g(x) &= f(x) + (-1)^n f^{(2n+2)}(x) \quad \text{but } f^{(2n+2)}(x) = 0 \\
&= f(x)
\end{aligned}$$

Hence

$$\begin{aligned}
\frac{d}{dx}\left[g'(x)\sin x - g(x)\cos x\right] &= f(x)\sin x \\
\int_0^{\pi} f(x)\sin x\, dx &= \int_0^{\pi} \frac{d}{dx}\left[g'(x)\sin x - g(x)\cos x\right] dx \\
&= \left[g'(x)\sin x - g(x)\cos x\right]_0^{\pi} \\
&= g(\pi) + g(0) \quad \text{which is an integer from part (ii)}
\end{aligned}$$

(iv) For $0 \leq x \leq \pi$ then

$$\begin{aligned}
x &\leq \frac{p}{q} \\
p - qx &\geq 0 \\
\Rightarrow f(x) &\geq 0 \\
f(x)\sin x &\geq 0 \quad \text{since } \sin x \geq 0 \quad \text{for } 0 \leq x \leq \pi \\
\int_0^\pi f(x)\sin x\,dx &\geq 0
\end{aligned}$$

Also $\sin x \leq 1$ for $0 \leq x \leq \pi$, so

$$\begin{aligned}
f(x)\sin x &\leq f(x) \\
&= \frac{x^n(p-qx)^n}{n!} \\
&\leq \frac{\pi^n p^n}{n!} \\
\int_0^\pi f(x)\sin x\,dx &\leq \frac{\pi^n p^n}{n!}\int_0^\pi dx \\
&= \frac{\pi^{n+1}p^n}{n!}
\end{aligned}$$

$$\therefore 0 \leq \int_0^\pi f(x)\sin x\,dx \leq \frac{\pi^{n+1}p^n}{n!}$$

(v) Given that $\pi p < m$ then

$$\begin{aligned}
\pi p &< m \\
\pi p &< m+1 \\
\pi p &< m+2 \\
&\vdots \\
\pi p &< m + (n-m-1) \\
\Rightarrow (\pi p)^{n-m} &< m(m+1)(m+2)\ldots(m+(n-m-1)) \\
\frac{(\pi p)^{n-m}}{n!} &< \frac{m(m+1)(m+2)\ldots(n-1)}{n!} \times \frac{(m-1)}{(m-1)!} \\
&= \frac{(n-1)!}{n!(m-1)!} \\
&= \frac{1}{n(m-1)!}
\end{aligned}$$

$$\therefore \frac{(\pi p)^n}{n!} < \frac{(\pi p)^m}{(m-1)!}\cdot\frac{1}{n}$$

(vi) From the results in part (iv) and part (v)

$$0 \leq \int_0^\pi f(x) \sin x \, dx < \frac{(\pi p)^m}{(m-1)!} \cdot \frac{1}{n}$$

This inequality holds true for **any** integer $n > 1$. This means that it should hold for large values of n.

However, note that as $n \to \infty$ then $\frac{(\pi p)^m}{(m-1)!} \cdot \frac{1}{n} \to 0$. This means that is possible for n to be sufficiently large such that $\frac{(\pi p)^m}{(m-1)!} \cdot \frac{1}{n} < 1$.

This means that there exists some values of n such that

$$0 \leq \int_0^\pi f(x) \sin x \, dx < 1.$$

This is a contradiction because $\int_0^\pi f(x) \sin x \, dx$ is an integer from (iii).

The initial assumption that π is rational is false. Hence, π is irrational.

Remark: Since $f(x) \sin x \geq 0$, it is only possible for $\int_0^\pi f(x) \sin x \, dx = 0$ if $f(x) \sin x = 0$ for all $0 \leq x \leq \pi$, which is obviously false so the left inequality can be written as a strict inequality.